

FICTIONAL CASE · SYNTHETIC DATA

CENTERLINE BANK · CLAUDE PILOT

AI the RMs actually use, and the CCO trusts

Built, not described · Measured, not promised · Compliant by design

Technical-challenge presentation · **Commercial Banking**

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Agenda

Roughly 45 minutes, then questions. Two solutions, run live, not described.

- 1 Who I am**
A 60-second intro
- 2 The situation**
3 RMs · \$350M · a shadow-IT catalyst · \$4 compliance · the data
- 3 How we built it**
Composable skills + MCP + server-side guardrails, in one architecture
- 4 Track A, Early-Warning** (Deliverable A)
Rebuild Tom's shadow workflow, run live
- 5 Track B, Relationship & Renewal** (Deliverable B)
Our creative find, run live
- 6 Trust, eval & adoption**
The honest 80/20, and the path to production

About me

SNAPSHOT



Franck Benichou

Senior AI Engineer · Carta (LPPA) · Toronto

- AI Engineering, Document Intelligence
- Data Science + Consulting / start-up: Intact · EY · Deloitte · Carta
- Technical reviewer · [DeepSeek in Practice](#) (Packt, 2025)
- Founder · HEC Forecast student data-science conference (2018 → present)
- Français · English · Español

WHY CAYLENT'S SENIOR FDE, AND WHY ME

- 1 **It's the work I already do.** AWS + Anthropic is my daily stack, the Anthropic APIs and Claude Code at Carta, in document intelligence.
- 2 **Why I'm excited.** I'm an AI builder who wants direct commercial impact on the front lines, prototyping and taking things into production.
- 3 **Why I'm a fit.** Beyond building, my unconventional, hybrid background brings judgement and client-facing strength, not just code.

The situation

Centerline's Commercial Banking division bought Claude for Work, and deployed nothing **yet**.

\$350M

portfolio managed by **3 relationship managers**

30–40%

of RM time on **documentation, covenant tracking & call prep**

0

workflows in production, **licenses bought, not used**

The CCO, **Rachel Torres**, has one stated priority: **early-warning** on contact gaps and financial deterioration, RM-private, compliance-gated. *My 90-day mandate.*

Meet the three RMs

Three RMs, three very different relationships with AI. Adoption is the real risk, so we design for each, not for the org chart.

Sarah Okonkwo

THE LINCHPIN · NATURAL TEACHER

Book: Meridian · BlueLine, both in covenant breach

Diligent logger. Burned once by an AI that gave **wrong covenant info**, so trust is existential.

If she adopts, she trains the team.

Tom DeLuca

THE BUILDER · POWER-USER

Book: Crestwood (A-rated) · Arcadia (construction)

Built 5 AI workflows on personal ChatGPT. Wants to **co-design**, not be policed.

Already proved AI helps, by breaking policy.

Marcus Webb

THE SKEPTIC · 22-YR VETERAN

Book: Summit, DSCR drifting toward the floor

Mentors 4 juniors. **~71 annual memos/yr.** Logs sparsely; trusts his own memory.

Win him and the rollout tips.

RM = Relationship Manager, a commercial bank's primary owner of a business-borrower relationship: monitors covenants and financial health, prepares periodic reviews and renewals, and is the client's main point of contact.

The catalyst: one RM didn't wait

IT AUDIT FINDING

Tom built 5 AI workflows on personal ChatGPT

Client financials and names in a personal account, **two compliance violations**. Five workflows: draw letters · memos · client emails · call-prep · weekly summary.

THE REFRAME

- **Right instinct, wrong tools.** Tom proved AI helps here, by breaking policy to do it.
- **Starting point, not blueprint.** We keep the instinct, fix the violations, and turn one person's hack into a governed, shareable library.

Compliance isn't a wrapper, it's the architecture

Section 4 is trickier than it looks. Five constraints decided every design choice that follows.

Section 4.2

No creditworthiness language, even implicitly. A disclaimer does **not** count.

Section 2.1

Credit grade / watchlist can't even be **inputs** to the AI.

Section 5

No AI on **Special-Assets** or litigation credits.

Section 4.1

Automated monitoring needs **CCO approval** to launch.

Section 4.3

Human review before anything is client-facing or filed.

The credit **judgment must originate with the RM**. Claude organizes facts and flags gaps, it never characterizes credit. And we enforce that **in code, not in a prompt**.

The signal's already there; the record can't be trusted

Eight files, current through **2025-05-31**. The hard part isn't finding data, the system of record is wrong three ways: **mis-dated logs** · **decisions that live only in email** · **status labels that lag the trend**.

Meridian

A careful action plan in the memo, that the data shows was **never executed**.

Crestwood

Attrition risk on a pristine, healthy borrower, invisible to every distress metric.

Across the book

Engagement **diverges from distress**, RMs go quiet on the credits that are sliding.

So the job isn't to find hidden data. It's to **organize scattered facts and make the record trustworthy**, and leave the judgment with the RM. *That's the spine of both demos.*

How I built it

Five decisions, and an honest answer to the brief's "is a single prompt the right architecture?"

LLM-driven workflows, simplest-first (Anthropic guidance)

Composable skill library

Small, single-purpose skills Claude selects and composes, not one mega-prompt.

MCP tool-use over copy-paste

Retrieval and computation are tools, which is also what makes Tom's workflow compliant.

Guardrails server-side

§2.1 / §4.2 enforced in the tools, they hold no matter what the model says, on any surface.

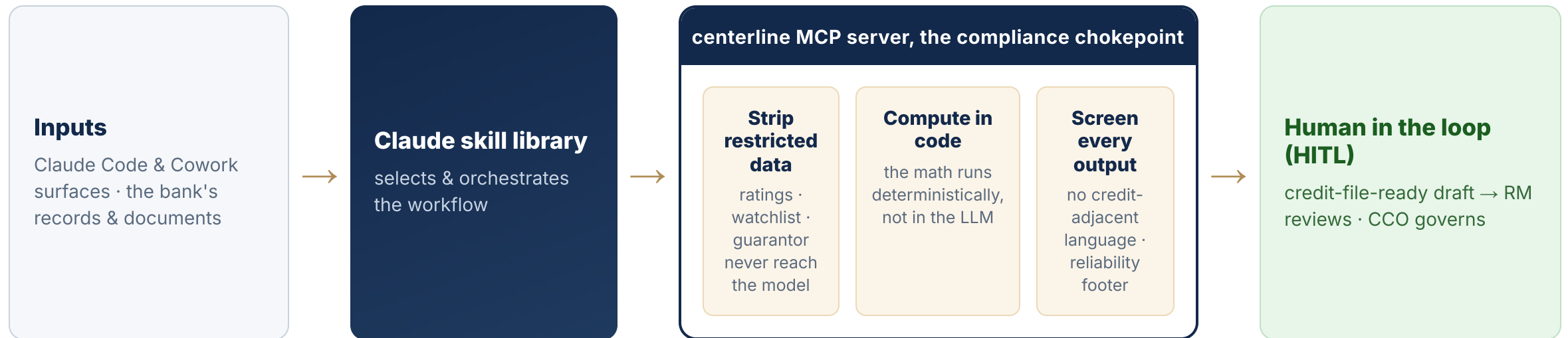
Build in Code → demo in Cowork

Cowork is the RM's no-terminal surface, the UI is the product, with no UI to build.

Single prompt or multi-step? The RM still types **one prompt**, but it now fires a multi-tool pipeline. Only the **memo (§4.2)** and **monitoring (§4.1)** need a genuine extra human / governance step.

Architecture in one picture

The AI organizes the facts; **people keep the judgment**, and every output passes through one compliance chokepoint.



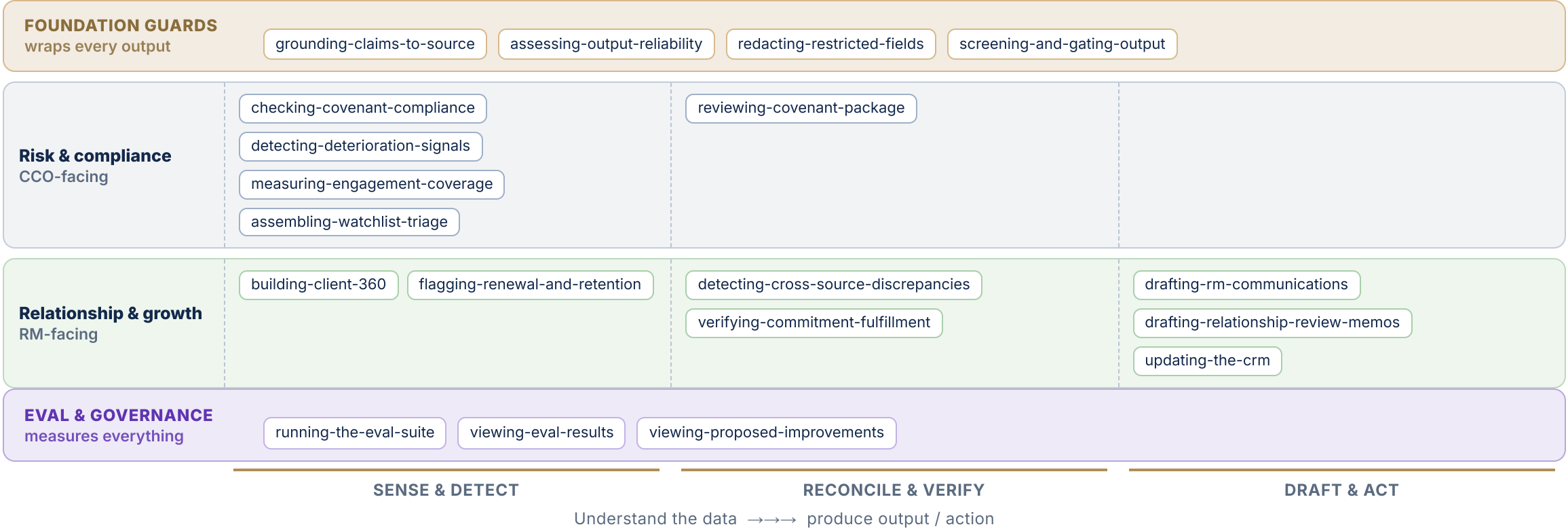
Packaged as one plugin, the same compliant workflows run in **Claude Code** and **Claude Cowork**, the RM's no-terminal surface.

Continuously evaluated, golden tests · live agent-eval · CI gate · reported to the CCO

THE COMPOSABLE SKILL LIBRARY

Skills, a capability map

The same library, mapped on two axes: **where it sits in the workflow** (understand → produce) and **who it serves** (risk / CCO vs relationship / RM).
Guards wrap the top; eval measures the bottom.



Fix Tom's flagged workflow

Tom's five shadow workflows, what's compliant as-is, and what genuinely needs re-architecting (the brief's "is one prompt right?").

Tom's workflow	Right architecture?	The compliance issue
Draw - response letter · client email · call - prep	✅ Keep as one prompt (retrieve, not paste)	§4.3 review · no credit bad - news
Relationship memo	❌ Break up: RM writes the assessment	§4.2: the AI may not author the credit judgment
Weekly portfolio summary (→ Track A)	❌ Break up: code computes the flags	§4.1: monitoring needs CCO sign - off
all five	n/a	§1 / §2 / §5: confidential data on a personal ChatGPT

Deliverable A rebuilds the weekly summary as the early-warning: **retrieve · deterministic flags · screen**, RM-private, §4.1-gated. The aggregation *was* the problem; moving the math into code is the fix.

"Who needs attention this week?"

PROMPT Who in my portfolio needs attention this week? Rank them and show me the facts behind each, don't characterize their credit, just the numbers and the gaps.

WHAT YOU'LL SEE

- Ranked by **risk × neglect**, Meridian #1, Crestwood last
- Every line **cited**, it ranks, it doesn't judge

UNDER THE HOOD

INPUT loan_performance · activity_log · portfolio_ref, all borrowers, §2.1-stripped

assembling-watchlist-triage → assemble_watchlist →

covenant · deterioration · engagement → screen_and_finalize

☒ skill ☐ MCP tool

OUTPUT ranked watchlist (cited) + Centerline PDF

The point: it ranks and cites, never characterizes credit (§4.2), RM-private. Crestwood sinks to the bottom: distress monitoring is blind to a *healthy* borrower, which sets up Track B.

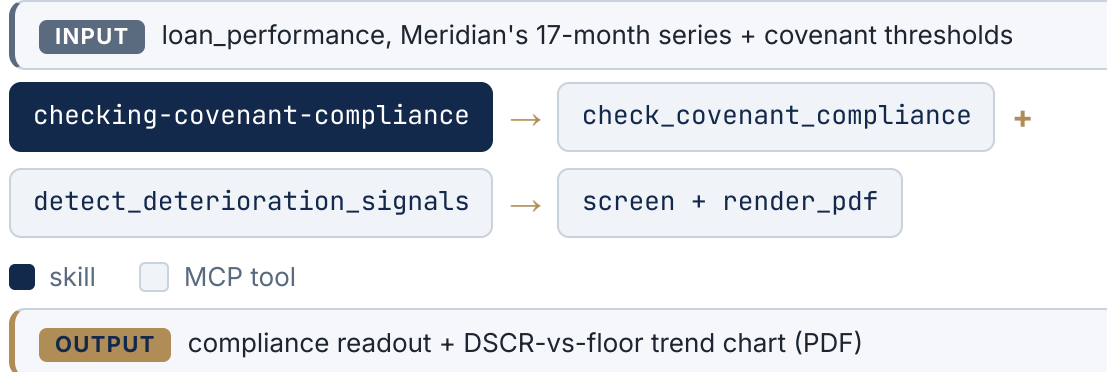
"Is Meridian compliant, cushion *and* trend?"

PROMPT Is Meridian covenant-compliant as of the latest month? Show DSCR and leverage against their covenant thresholds with the cushion, the revolver level, and whether the reported status matches the trend.

WHAT YOU'LL SEE

- DSCR **1.03 vs 1.20** (breach) · leverage 4.5 · revolver **85% vs the 75% cap**
- The status field read **"Compliant"** while DSCR slid 1.45 → 1.03

UNDER THE HOOD



The point: the math runs **in code, not the LLM**. The field said "fine"; the trend said "sliding." The AI shows both as facts, the RM owns the call.

"Who have I gone quiet on?"

PROMPT Which of my distressed borrowers have I actually gone quiet on? Count days since the last real two-way contact, a one-way notice or a missed call doesn't count.

WHAT YOU'LL SEE

- Meridian: **78 days** of substantive silence (as of 2025-05-31)
- Naive log reads 33, a one-way notice isn't contact → **undercounts by 45 days**
- A risk **no single field flags**, RM-private & dismissible

UNDER THE HOOD

INPUT activity_log + emails, Meridian (two-way contact only)

measuring-engagement-coverage

→ measure_engagement_coverage →

screen + render_pdf

☒ skill ☐ MCP tool

OUTPUT engagement-gap readout + chart (PDF)

Flag ≠ action. Tom's own weekly summary flagged the Crestwood gap, and he still lost momentum. Detection isn't enough → that's why **Track B** is about the relationship and the record.

Our own find: Relationship & Renewal

The use case I found by reading the data, not on Tom's list, and not the obvious pain point.

The creative claim

- **Close-the-loop**, verify the bank kept *its own* commitments
- **Cross-source reconciliation**, emails ↔ CRM ↔ structured data
- **Proactive retention**, catch the healthy client before they shop

Combines sources no one reconciles, a risk Tom didn't see.

Compliant cleanup of Tom's other workflows

- Relationship memo · covenant-package intake · overdue-docs email
- The brief's "avoid" list, rebuilt compliant
- **Explicitly not** the creative claim

Said out loud: the brief says don't headline the memo, covenant-tracking, or overdue emails, so those are the **compliant cleanup**.
The find is **close-the-loop + reconciliation + retention**, demoed across all three RMs.

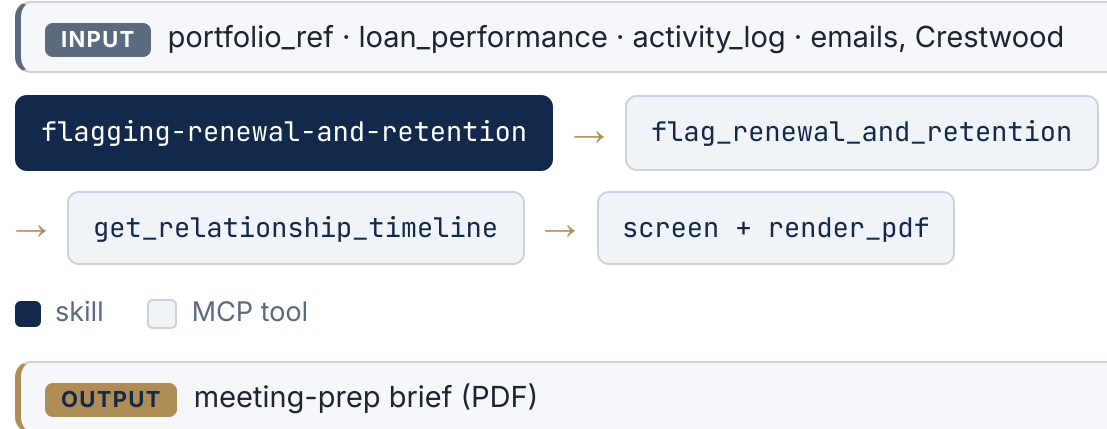
"Prep me for my Crestwood meeting"

PROMPT Prep me for my Crestwood meeting, pull the relationship picture, the renewal/maturity clock, and the competitive situation; flag anything I should act on before they shop us.

WHAT YOU'LL SEE

- The **inversion**: the one you lose by being *slow*, not by credit risk, pristine metrics
- Matures 2026-08-31 (**457 days**) · First Midwest **4.90%** vs your 5.05% · pricing-exception deadline passed · no renewal memo

UNDER THE HOOD



The point: facts and "engage", **never a rate** (\$4.2; pricing is the committee's call). The *healthy* borrower A1 is structurally blind to.

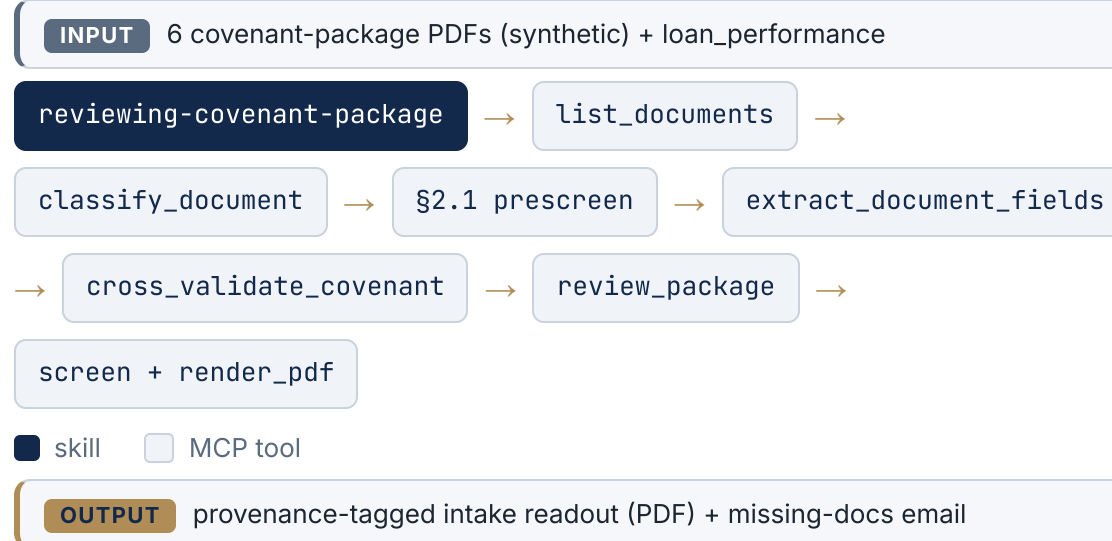
"What's in the package, and do the ratios reconcile?"

PROMPT Sarah's Meridian Q1 covenant package just arrived in data/synthetic/meridian-package/. Review it, what's in it, what's missing, and do the ratios reconcile?

WHAT YOU'LL SEE

- §2.1 **guarantor financials refused on intake** · ACORD insurance cert → **other**, skipped
- Certified **1.23 / 3.76** vs recomputed **1.03 / 4.50**, gap = **\$2.0M EBITDA add-backs**; the bank's own data agrees with the recompute
- Footer reads **Partial**, each reason tagged **[tool · source]**.
Then: "Draft the missing-docs email" (wf3)

UNDER THE HOOD



Sarah's scar: every figure cites its document + field + tool; the §2.1 refusal is **enforced, not a disclaimer**; whether the §1.01 add-backs are adequately supported is **the RM's call, not mine** (§4.2).

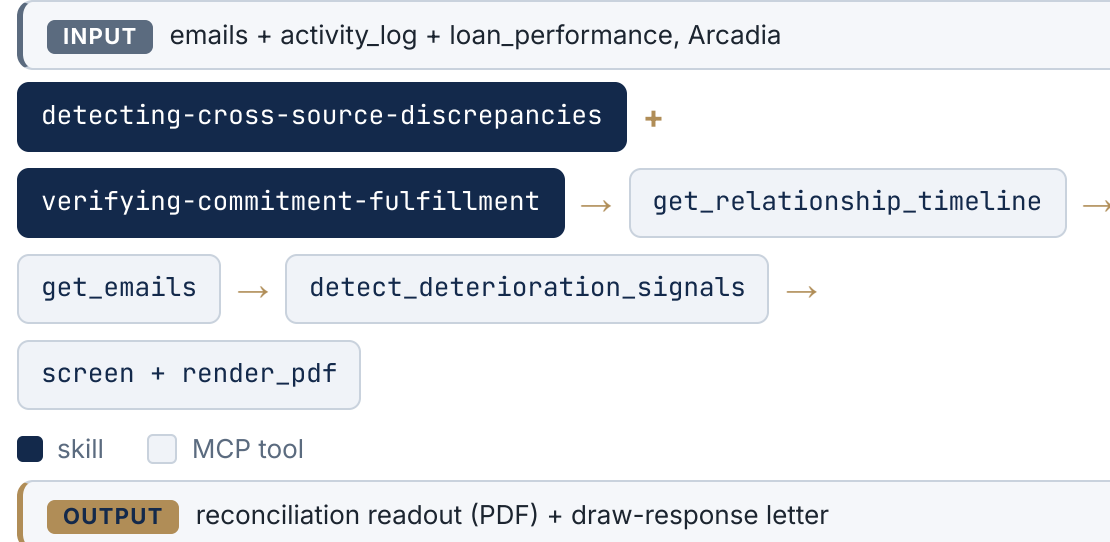
"Reconcile the emails vs the CRM log"

PROMPT Reconcile Arcadia's email thread against the CRM log, what actually happened, what's mis-dated, what decision isn't in the file, and what's still open?

WHAT YOU'LL SEE

- The Apr-09 CRM row sits **~2 weeks before** the real Apr-22→25 thread, it poisons contact-recency
- Marcus Chen's **"don't waive the 75%"** ruling lives only in email → surfaced **verbatim + attributed**, with a **propose-only** CRM entry
- Open: 71% < 75%, Friday docs, names withheld. **Then:** "Draft the draw-response letter" (wf1)

UNDER THE HOOD



The \$4.2 moment: the AI **transcribes the credit officer's decision into the file, it never makes one.** It combines sources no one reconciles, and surfaces the bank's own dropped follow-through.

"Build my Summit annual relationship-review memo"

PROMPT Build my Summit annual relationship-review memo.

WHAT YOU'LL SEE

- AI **assembles + cites** every factual section (17-mo DSCR arc, covenant table, engagement, open items)...
- ...then **II STOPS** and asks Marcus for his assessment **in his own words**, the AI never writes the judgment
- Drafts the narrative **around his verbatim words**. Removes ~71 memos/yr without changing how he thinks

UNDER THE HOOD

INPUT loan_performance · activity_log · emails · portfolio_ref, Summit

drafting-relationship-review-memos →

assemble + cite (client-360) →

II RM writes the assessment →

draft → screen + render_pdf

■ skill □ MCP tool □ human step

OUTPUT annual memo (markdown + PDF)

II THE HUMAN STEP, WHAT MARCUS FILLS IN (VERBATIM, ATTRIBUTED)

"Coverage thinned to 1.12x, just over the 1.10x floor, but I read it as seasonal softness, not a structural decline; leverage at 3.6x leaves cushion. My real concern is commercial, not credit: the Location-3 renewal at an 18% bump, and Castillo's sounded more stressed lately. Eyes on the Q2 package before I move my view."

Outlook **Watch / monitor** · Next steps **follow up on the Location-3 lease; request updated Q2 projections** · Filed under **Marcus Webb (RM of record)**

§4.2, made literal: the AI assembles the 80%; Marcus owns the credit judgment it never writes. The skeptic's wedge.

How the CCO can trust it, with closed-loop learning

Reliability is a **reported number across both tracks**, not a vibe. Four eval layers, plus a CI prod-gate.

Layer 1

Deterministic golden set

95/95 source-grounded cases across A1–A3 + B1–B4. Tests the **tool code**, the LLM is never invoked.

Layer 2

Observability

Per-prompt scorecards, expected-vs-actual steps, guard hits, reliability footer, latency, across every demo prompt.

Layer 3

Live agent-eval

Runs the prompts through Claude headless and grades the **model's own decisions**, tool choice, §4.2 on its narration, fact faithfulness.

Layer 4

Improvement loop

Advisory only, proposes skill edits, can't touch guards / core / eval keys. A human applies them.

AI proposes, a human disposes (§4.3). The loop caught a real §4.2 footer issue, left unapplied as a live governance artifact. A **CI prod-merge gate** (black · flake8 · deterministic suites) blocks any failing merge. *Layer 1 tests the tools; Layer 3 tests the model, don't conflate them.*

"How is it doing, and what should we change?"

RUN LIVE "Show me the latest agent eval."
"Show me the latest proposed improvements to our skills and automations."

WHAT YOU'LL SEE

- **Eval scorecard**, golden 95/95 + the live agent-eval, across both tracks
- **Improvement proposals**, incl. the §4.2 footer finding the loop caught, **left unapplied** for a human to approve

UNDER THE HOOD

INPUT committed eval + improvement reports (pulled fresh, cross-surface)

viewing-eval-results + viewing-proposed-improvements →

get_latest_report

☒ skill ☐ MCP tool

OUTPUT the scorecard + the proposals, shown in Cowork

The loop drives the work: it independently caught the §4.2 issue and proposed the exact fix, we left it unapplied. **AI proposes, a human disposes (§4.3)**, the CCO's audit trail.

HONEST EVALUATION

What it does well, and the hard 20%

✓ What works

- Grounded & **cited**; compliant **by construction**
- The math is **deterministic**, not the LLM
- Reusable across the book; both creative use cases run on **real data**

The hard 20%

- **Flag ≠ action**, Tom flagged Crestwood and still lost momentum
- Value is **gated by logging discipline** (Marcus logs sparsely)
- Track A is **blind** to healthy & construction borrowers, by design
- Doc-intel runs on **synthetic** documents (clearly labeled)

This is the honest 80/20, what works, and where it falls short. And **none of the creative claim rests on fabricated data**: both creative use cases run on the real files.

ADOPTION: THE REAL RISK

No single tool wins all three RMs: a portfolio does

Sarah

START HERE · SHE TRAINS THE TEAM

Her trust scar → grounding; her surveillance fear → RM-private design. The "10-minute Monday."

Wins on: A1-A3 + B2

Tom

THE CO-BUILDER

Shadow workflow → sanctioned & shareable (\$6 amnesty). Make him the partner, not the cautionary tale.

Wins on: B1 + B3

Marcus

THE SKEPTIC · MAKE-OR-BREAK

The memo removes ~71/yr **without changing how he thinks**, \$4.2 hands him the pen.

Wins on: B4

Adoption is the CCO's stated #3 risk. **Start with Sarah** (she trains the floor), **Marcus** is the proof point, **Tom** co-authors, a portfolio, not one tool.

PRODUCTION

From pilot to production

Cowork = the RM surface

One plugin, no terminal, no UI to build, the same compliant workflows the RMs already saw.

Monitoring · \$4.1

Presented as designed, RM-private, **pending CCO sign-off**, fast-trackable.

Residency · \$2.3

Local scheduled jobs > cloud routines, the bank's data stays put.

Scheduled jobs

A Sunday-night pre-compute → Sarah's watchlist + open items waiting Monday. Prepare-only, never auto-act.

Query observability & cost

Every query + response logged and **scanned for non-compliant (\$4.2) language**, not just artifacts; per-RM/skill **token & spend budgets** with caps + alerts. OTel → Datadog.

Sandboxing & least-privilege MCP connectors

The MCP server reaches real data through **least-privilege, read-only connectors**; skills run sandboxed, defense beyond the server chokepoint.

PRODUCTION PATH ON AWS Our local MCP + guards map 1:1 onto **Bedrock AgentCore Gateway** (managed MCP · least-privilege) · **Runtime** (microVM isolation) · **Observability** (token · OTel → CloudWatch / Datadog) · **Bedrock Guardrails**, defense-in-depth on our deterministic guards.

What 90 days delivers: pilot both tracks with the three RMs → CCO sign-off on monitoring → a governed data store + funded API tier for rollout.

Built. Trustworthy. Adopted.

Two solutions, run live: compliant by construction, geared for production.

A: kept Tom's instinct; fixed the violations.

B: the AI organizes the facts; the RM keeps the judgment.

C: adoption is the real risk; a portfolio, each tool earning its RM.

Questions?

Franck Benichou · [linkedin.com/in/franckbenichoudata](https://www.linkedin.com/in/franckbenichoudata) · github.com/benichou/centerline-claude-pilot

What each skill achieves

FOUNDATION / TRUST (CROSS-CUTTING)

grounding-claims-to-source · cite every claim to its source; uncitable → UNVERIFIED.

assessing-output-reliability · qualitative reliability footer (Grounded / Partial), never a %.

redacting-restricted-fields · §2.1 inputs (rating / watchlist / guarantor) never reach the model.

screening-and-gating-output · block §4.2 credit language; tag §4.3 review.

RISK & EARLY-WARNING (TRACK A)

checking-covenant-compliance · latest-month DSCR/leverage vs thresholds, with cushion.

detecting-deterioration-signals · full-history trends, status mislabel, thin cushion, lifecycle.

measuring-engagement-coverage · days since the last substantive two-way contact.

assembling-watchlist-triage · rank the book by risk × neglect (a facts order, not a rating).

RELATIONSHIP & RECONCILIATION (TRACK B)

flagging-renewal-and-retention · healthy borrower at attrition risk; "engage," never a rate.

detecting-cross-source-discrepancies · mis-dated entries, email-only decisions, contradictions.

verifying-commitment-fulfillment · commitments met / unmet / open / unverifiable.

DOCUMENT INTELLIGENCE (TRACK B)

reviewing-covenant-package · classify, refuse restricted, extract, recompute, completeness.

CONTEXT & SYNTHESIS

building-client-360 · consolidated dossier; flags the dimensions not covered.

DRAFTING / OUTPUT

drafting-rm-communications · factual client message; no commitments / credit bad-news.

drafting-relationship-review-memos · assemble facts, RM writes the assessment (§4.2 pause).

CRM ENRICHMENT

updating-the-crm · structured note + tasks, propose-only; RM approves the write.

EVAL & GOVERNANCE

running-the-eval-suite · run the golden set on live tools; refresh the scorecard.

viewing-eval-results · display the latest scorecard / agent-eval (read-only).

viewing-proposed-improvements · display the loop's advisory proposals (read-only).

The centerline MCP server + its tools

A local Python **MCP server** exposing the bank's data and the deterministic analysis / compliance logic as callable tools, the project's **cross-surface compliance chokepoint**: every retrieval strips restricted inputs (§2.1), halts prohibited borrowers (§5), and refuses guarantor documents; every artifact can route through the §4.2 output screen, **in code, not a prompt**. Runs host-side via `uv`, loaded by Claude Code (`.mcp.json`) and bridged into Claude Cowork (`claude_desktop_config.json` , type: `sdk`).

RETRIEVAL (§2.1 / §5 ENFORCED)
get_borrower_dossier profile + latest performance
get_loan_performance monthly DSCR / leverage / revolver
get_activity_log the CRM contact log
get_emails the email thread
get_relationship_timeline log + emails, one timeline
list_documents enumerate a package, host-side

TRACK-A ANALYSIS (DETERMINISTIC)
check_covenant_compliance snapshot vs thresholds + cushion
detect_deterioration_signals trend, mislabel, lifecycle
measure_engagement_coverage substantive-contact gap
assemble_watchlist rank by risk × neglect
flag_renewal_and_retention inverse radar; never a rate

DOCUMENT INTELLIGENCE
classify_document type a PDF; §2.1 pre-screen
extract_document_fields per-type schema extraction
cross_validate_covenant certified vs recomputed vs bank
review_package completeness + quality flags

COMPLIANCE & OUTPUT
screen_and_finalize §4.2 scan + reliability footer + §4.3 tag
render_pdf credit-file-grade PDF, letterhead

EVAL & GOVERNANCE
run_evals golden set on live tools; refresh scorecard
get_latest_report pull + show eval / improve / observability